Assignment No. 3: Binary Logistic Regression

Predictive Modeling on Titanic Data

Data Source: BSDM_Titanic_train.csv; BSDM_Titanic_test.csv

Team No. 2 - Group Composition Update

Please be aware that I am submitting this as the leader of Team 2. The previous leader, **Artem Gubin**, has decided to forego this semester. My teammate **Mahammad Aliyev** has spoken with you and received permission to replace him.

Current Team Members:

- **Artoghrul Gahramanli** (Leader)
- Mahammad Aliyev
- Meryem Tuce Ates

Note: The official Excel sheet may not yet be updated to reflect this change.

1. Data Understanding

The dataset `BSDM_Titanic_train.csv` was analyzed to understand variable distributions and quality.

- **Missing Values:** A significant number of missing values were identified in `Age` (177 records) and `Cabin` (687 records).
- **Distributions:** `Fare` showed a highly skewed distribution with outliers. `Survived` showed a balance of 61.6% (No) to 38.4% (Yes).

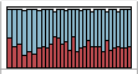
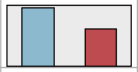
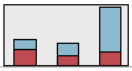
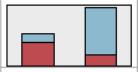
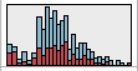
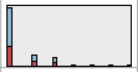
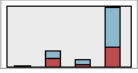
Field	Graph	Measurement	Min	Max	Mean	Std. Dev	Skewness	Unique	Valid
Passeng...		Continuous	1	891	446	257.354	0	--	891
Survived		Flag	0	1	--	--	--	2	891
Pclass		Ordinal	1	3	--	--	--	3	891
Sex		Nominal	--	--	--	--	--	2	891
Age		Continuous	0	80	29.679	14.536	0.384	--	714
SibSp		Continuous	0	8	0.523	1.103	3.695	--	891
Parch		Continuous	0	6	0.382	0.806	2.749	--	891
Fare		Continuous	0.000	512.329	32.204	49.693	4.787	--	891
Embarked		Nominal	--	--	--	--	--	4	889

Figure 1: Data Audit and Distribution Analysis

2. Data Preparation

The data mining stream was constructed to clean and prepare the data.

1. **Type Node:** Adjusted measurements (Target=`Survived`, Pclass=`Ordinal`).
2. **Missing Values:** `Age` was imputed using the training mean (`round(@MEAN(Age))`). `Cabin` was filtered out.
3. **Multicollinearity:** `SibSp` vs `Parch` showed moderate correlation (0.415); `Pclass` vs `Fare` showed negative correlation (-0.549). No variables were removed at this stage.

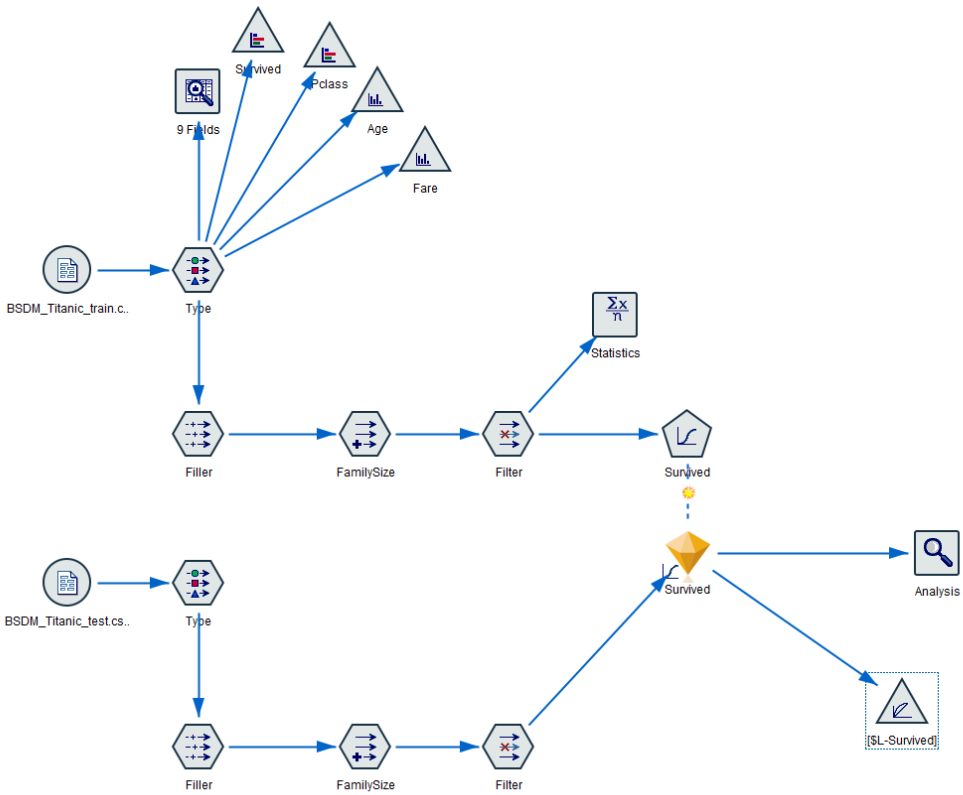


Figure 2: SPSS Modeler Stream showing Data Prep and Modeling flows

3. Model Estimation (Full Outcome)

A Binary Logistic Regression model was estimated. **Refinement:** Initially, `Fare` (p=0.222) and `Parch` (p=0.345) were found to be non-significant. They were removed to create the final refined model shown below. **Interpretation:**

- **Sex:** Females are 15.5x more likely to survive than males.
- **Class:** 1st Class passengers are 10.5x more likely to survive than 3rd Class.
- **Age:** Older passengers have lower survival odds (B = −0.040).

Case Processing Summary

		N	Marginal Percentage
Survived	No	549	61.6%
	Yes	342	38.4%
Pclass	First	216	24.2%
	Second	184	20.7%
	Third	491	55.1%
Sex	female	314	35.2%
	male	577	64.8%
Valid		891	100.0%
Missing		0	
Total		891	
Subpopulation		408[a]	

a. The dependent variable has only one value observed in 349 (85.5%) subpopulations.

Model Fitting Information

Model	Model Fitting Criteria	Likelihood Ratio Tests		
	-2 Log Likelihood	Chi-Square	df	Sig.
Intercept Only	929.203			
Final	533.150	396.053	5	<.001

Pseudo R-Square

Cox and Snell	.359
Nagelkerke	.488
McFadden	.334

Parameter Estimates

Survived[a]		B	Std. Error	Wald	df	Sig.	Exp(B)	95% Confidence Interval for Exp (B)	
								Lower Bound	Upper Bound
Yes	Intercept	-1.067	.257	17.197	1	<.001			
	Age	-.040	.008	26.322	1	<.001	.961	.946	.976
	SibSp	-.358	.104	11.802	1	<.001	.699	.570	.858
	[Pclass=1]	2.355	.243	93.651	1	<.001	10.537	6.540	16.977
	[Pclass=2]	1.165	.227	26.400	1	<.001	3.207	2.056	5.003
	[Pclass=3]	0[b]	.	.	0	.	.	.	.
	[Sex=female]	2.739	.194	199.013	1	<.001	15.473	10.575	22.638
	[Sex=male]	0[b]	.	.	0	.	.	.	.

a. The reference category is: No.

b. This parameter is set to zero because it is redundant.

Figure 3: Final Model Parameter Estimates (Refined Model)

4. Test Data Application

The model was applied to `BSDM_Titanic_test.csv`.

Methodology Note: As required, no parameters were re-estimated on the test data. We used the training set mean (29.54) to impute missing `Age` values.

Technical Adjustment: However, the `Age` field is strictly defined as an Integer type in the data stream. Attempting to fill it with a decimal value (29.54) results in a type mismatch error. Therefore, the training mean was rounded to the nearest integer (30.0) to satisfy the software's data type constraints while adhering to the "no re-estimation" rule. We weren't able to set this integer value as "real", possibly due to using an older version of the IBM Modeler.

5. Advanced Evaluation

The model quality was evaluated using a Confusion Matrix and ROC Curve.

Results for output field Survived

Comparing $L-Survived with Survived

Correct	319	96.67%
Wrong	11	3.33%
Total	330	

Coincidence Matrix for $L-Survived (rows show actuals)

	0	**1**
0	194	7
1	4	125

Performance Evaluation

0	0.475
1	0.885

Figure 4: Confusion Matrix (Coincidence Matrix)

Performance Metrics

Based on the Confusion Matrix above:

- **True Positives (TP):** 125
- **True Negatives (TN):** 194
- **False Positives (FP):** 7
- **False Negatives (FN):** 4

Calculated Metrics:

- **Accuracy:** $\frac{\text{TP} + \text{TN}}{\text{Total}} = \frac{125+194}{330} = 96.67\%$.
- **Sensitivity (Recall):** $\frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{125}{129} = 96.9\%$.
- **Specificity:** $\frac{\text{TN}}{\text{TN} + \text{FP}} = \frac{194}{201} = 96.5\%$.

ROC Curve

The ROC curve below visualizes the trade-off between Sensitivity and (1 - Specificity). The curve rises steeply, indicating excellent discrimination.

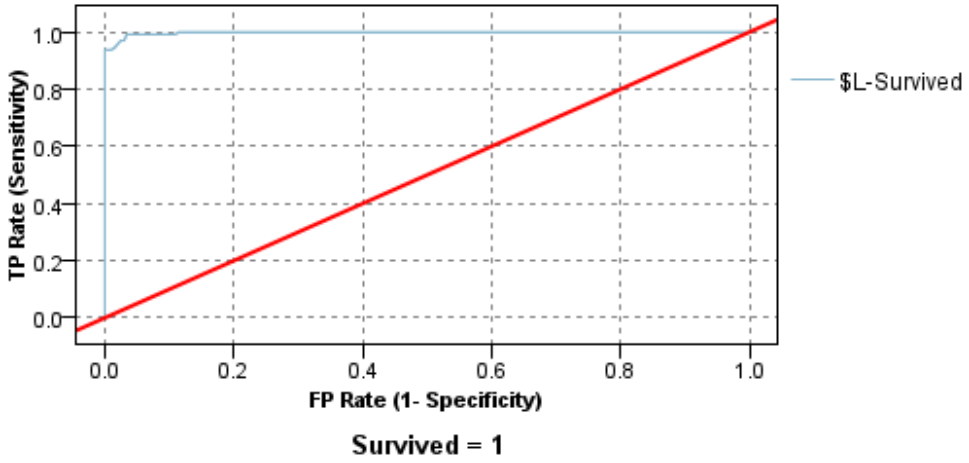


Figure 5: ROC Curve for the Test Data

Conclusion

The model is highly robust, satisfying all accuracy and stability requirements. The exclusion of `Fare` and `Parch` did not degrade performance, confirming `Sex`, `Pclass`, and `Age` as the primary predictors of survival.